MTH2021 Mathematical Methods 2

Homework set 4

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1. Sketch the pizza-wedge shaped closed contour C consisting of the segments:

   i. C_1: $z = r\exp(\mathrm{i}\alpha)$, $0 \le r \le R$
   ii. C_2: $z = R\exp(\mathrm{i}t)$, $\alpha \le t \le \beta$
   iii. C_3: $z = r\exp(\mathrm{i}\beta)$, $R \ge r \ge 0$

   with $0 \le \alpha < \beta \le 2\pi$.

   Let n be a positive integer.

   (a) Consider the analytic function $f(z) = z^n$. Verify that $f(z)$ integrates to zero around C by evaluating explicitly its integrals along C_i, $i = 1, 2, 3$.

   (b) Consider $f(z) = z^{*n}$. By a method of your choice, show that $f(z)$ is nowhere analytic.

   Evaluate and discuss the integral of $f(z)$ around C above, as a function of n, α and β.

2. Let C be the contour $|z| = 3$. Use Cauchy's theorem or the Cauchy integral formula as appropriate to evaluate the following integrals:

$$\text{(a)}\quad \oint_C \exp(2z)\,dz \qquad \text{(b)}\quad \oint_C \frac{\exp(2z)}{z+4}\,dz \qquad \text{(c)}\quad \oint_C \frac{\exp(2z)}{(z+\mathrm{i})^3}\,dz$$

3. Let the contour C be the square of side 2, centre at the origin and sides parallel to the x- and y-axes. Use Cauchy's theorem or the Cauchy integral formula as appropriate to evaluate the following integrals:

$$\text{(a)}\quad \oint_C \frac{\sinh z}{z-\mathrm{i}/2}\,dz \qquad \text{(b)}\quad \oint_C \frac{\cos(z+1)}{z^4}\,dz \qquad \text{(c)}\quad \oint_C \frac{\sinh z}{(z^2-4)^2}\,dz$$

4. By applying the Cauchy integral formula for the n-th derivative to the function $\exp(-\mathrm{i}z)$ and the contour $|z| = r$ show that

$$\int_0^{2\pi} \exp(r\sin\theta)\,\cos(n\theta + r\cos\theta)\,d\theta = \frac{2\pi r^n}{n!}\cos(n\pi/2)$$
$$\int_0^{2\pi} \exp(r\sin\theta)\,\sin(n\theta + r\cos\theta)\,d\theta = \frac{2\pi r^n}{n!}\sin(n\pi/2)$$

5. Find the first 4 terms of the Taylor series, around the point $z = 0$, of the functions

$$\text{(a)}\quad f(z) = z\cos z \qquad\qquad \text{(b)}\quad f(z) = \frac{\exp z}{1+z}$$

6. Find the Laurent series of the following functions in the specified region:

(a)

$$f(z) = \frac{1}{(z+1)(z-1)} \qquad \text{for} \qquad 0 < |z-1| < 2$$

(b)

$$f(z) = \frac{1}{(z+1)(z-1)} \qquad \text{for} \qquad |z| > 1$$

(c)

$$f(z) = \frac{1}{z(z-1)} \qquad \text{for} \qquad |z| > 1$$

(d)

$$f(z) = \frac{\sin z}{z^3} \qquad \text{for} \qquad |z| > 0$$